

Result

The theoretical in-degree probability distribution for nodes of type x (where we numerically investigate the case $x \in \{a, b\}$) in the limit $\lim_{t \rightarrow \infty} p(k, t)$ for a homophilic Price model is given by:

$$p_{\infty, x}(k) = \begin{cases} p_{0, x} & k = 0 \\ p_{0, x} \cdot \frac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)} & k > 0 \end{cases} \quad (1)$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ where denotes the beta function. While the underlying network model is defined only for integer-valued degrees, corresponding to $k \in \mathbb{N}$, the analytical expression admits a natural extension via analytic continuation to $k \in \mathbb{R}$. This extension is used solely for the purpose of figure generation and visual comparison with the theoretical distributions.

Parameters

The distribution parameters are:

$$\alpha_x = \frac{\mu_x}{\lambda_x} \quad (2)$$

$$\gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x} \quad (3)$$

$$p_{0, x} = \frac{1}{1 + \mu_x \tilde{Z}} \quad (4)$$

where:

- μ_x is the initial attractiveness parameter for type x nodes
- λ_x is the coupling strength for type x nodes
- $\lambda_x = f_x h_x + (1 - f_x)(1 - h_x)$
- f_x is the probability a new node added to the system is of type x .
- For simplicity h_x is the probability of a connection between nodes of the same type. Consequently, the probability of a connection between different type nodes is $1 - h_x$
- $\tilde{Z} = \frac{m}{Z}$ where $Z = g_a \lambda_a + g_b \lambda_b + f_a \mu_a + f_b \mu_b$ is the partition function
- $Z(t) = Z \cdot t$ is the time-dependent partition function
- m is the number of edges added per timestep
- g_x is the asymptotic mean in-degree for type x nodes

1 Definitions not yet in text

- Preferential attachment
- Homophilic Preferential attachment

2 Graph Theory Intro (unfinished)

Abandoned intro part on motivating this research: Suppose the fraction of male AI researchers is $f \in [0, 1]$. There is an affect in the social sciences... this can be modeling using node homophily...

Airlifted from lit review to give starting point to graphs part: A graph, the mathematical object whose real-world instantiation is called a network, $G = (V, E)$, consists of a node set V and an edge set E . We want to study a type of graph called a directed graph. In directed graphs, edges are ordered pairs; $(u, v) \neq (v, u)$ of nodes, $u, v \in V$.

Consider a network containing N nodes. This system can be thought of as containing two types of node: an x type node, and nodes that are not type x . It follows therefore,

$$N = N_x + N_{x'} \quad (5)$$

where N_x is the number of type x nodes and $N_{x'}$ is the number of nodes that are not type x .

We can connect these nodes by drawing arrows from a given source node to a given target node. We call these arrows directed edges. The in-degree of a given node is the total number of arrows pointing toward this node, $k^{(\text{in})}$. The out-degree of the node, $k^{(\text{out})}$, is the number of outward arrows originating from the node.

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3 Scale-free power laws and the philosophy of system modeling (not started)

Section for providing the outline to some network results and stuff other 4th year physics students will need outlined: This section will address and therefore provide the appropriate background on why we chooseto model our system as using a homophilic preferential attachment model. Discuss the Albert-L'aszl'o Barab'asi and R'eka Albert paper. Disuss some general ideas on modeling real-world phenomena using networks.

4 Theoretical Result Derivation

Consider a directed network composed of nodes of different types. In particular, the network contains nodes of type x and nodes not of type x , denoted x' . Let $n_x(k, t)$ be the number of x type nodes with an in-degree k at the time t . The expected number of x type nodes with an in-degree k at time $t + 1$ can be expressed as:

$$n_x(k, t+1) = n_x(k, t) + m\Pi_x(k-1, t)n_x(k-1, t) - m\Pi_x(k, t)n_x(k, t) + \delta_{k,0} \quad (6)$$

The term,

$$m\Pi_x(k-1, t)n_x(k-1, t) \quad (7)$$

is the expected number of nodes that transition from degree $k-1$ to degree k in a single timestep. Since there are m new edges added by the new node created in a timestep, there are m opportunities for a given x type node to receive a new in-edge. $\Pi_x(k-1, t)$ is the probability that a type x node with degree $k-1$ receives a new in-edge. We omit consideration of the same node gaining two or more in-degrees in a single timestep here since these so-called multiedge terms are an $O(m\Pi)$ correction to our calculation, which in the long time limit (the limit as $t \rightarrow \infty$), is negligible [1]. The quantity $n_x(k-1, t)$ is the number of these x type nodes with in-degree $k-1$.

It therefore follows, by an identical line of reasoning, that the term,

$$m\Pi_x(k, t)n_x(k, t) \quad (8)$$

represents the expected number of nodes leaving degree k by gaining an edge and moving to degree $k+1$.

Finally, $\delta_{k,0}$ is a Kronecker delta that accounts for the case that, given the newly generated node is x type, it will be an addition to the $n(k=0, t)$ count, as the new node will necessarily have an in-degree of 0.

Now, the $\Pi_x(k, t)$ term itself can be expressed as,

$$\Pi_x(k, t) = \frac{f_x h_x k + (1 - f_x)(1 - h_x)k + \mu_x}{Z(t)} \quad (9)$$

which is a functional form known to give rise to a scale-free power-law degree distribution, a property known to be common across many large networks [2, 3]. f_x denotes the probability that a newly added node is of type x . Since each node is either of type x or not, the probability that a node is not of type x is $1 - f_x$. h_x denotes a weighting preference that a source node forms a connection with a node of the same type, $h_x \in [0, 1]$. This parameter allows us to capture, and exogenously tune, the degree of homophily in the model. For simplicity,

we assign the weighting to a connection between nodes of different types to be $1 - h_x$. This constitutes a modelling simplification, as it assumes symmetry between self-attachment and cross-type attachment probabilities. This means the homophily parameters, which we can represent by a 2×2 matrix, where the columns i correspond to the type of the newly added node and the rows j to the possible target node types, reduces to the following,

$$h_{ij} = \begin{bmatrix} h_{xx} & h_{xx'} \\ h_{xx'} & h_{x'x'} \end{bmatrix} = \begin{bmatrix} h_x & 1 - h_x \\ 1 - h_x & h_x \end{bmatrix} \quad (10)$$

The product $f_x h_x$ therefore represents the expected weighting in the preferential attachment process for same-type nodes of in-degree k to acquire a new edge, while $(1 - f_x)(1 - h_x)$ corresponds to the analogous weighting for different-type nodes of in-degree k . The consequences of repeating this calculation with this assumption removed are left for future work. The constant, $\mu_x \in \mathbb{R}^+$ is an exogenous parameter that controls the initial attractiveness of type x nodes, allowing us to study how asymmetric initialisation affects network evolution. It is useful to recognise that this numerator can be rewritten as,

$$\Pi_x(k, t) = \frac{\lambda_x k + \mu_x}{Z(t)} \quad (11)$$

where we have factorised the k in-degree terms and labeled the resulting factor λ_x ($\lambda_x \in \mathbb{R}^+$). The $Z(t)$ denominator is the normalization factor. It ensures that the probabilities of all existing nodes being targeted by the new directed edge sum to 1.

$$\sum_{i \in V_x} \Pi_{x,i} + \sum_{i \in V_{x'}} \Pi_{x',i} = 1 \quad (12)$$

which can be written as,

$$\sum_{i \in V_x} \frac{\lambda_x k + \mu_x}{Z(t)} + \sum_{i \in V_{x'}} \frac{\lambda_{x'} k + \mu_{x'}}{Z(t)} = 1 \quad (13)$$

Using this constraint, we can begin to solve for $Z(t)$. If we work under the conjecture that as $t \rightarrow \infty$ both the proportion of in-edges (where number of in-edges for an x type is $E_x(t)$) and fraction of node types in the model for a given node type tend to constant values, i.e.,

$$\lim_{t \rightarrow \infty} E_x(t) = \langle g_x \rangle t \quad (14)$$

where $\langle g_x \rangle, \langle g_{x'} \rangle \in \mathbb{R}^+$. and,

$$\lim_{t \rightarrow \infty} N_x = f_x t \quad (15)$$

we can now rearrange (11) for $Z(t)$ in the long time limit. To see this, let us focus on,

$$\sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} \quad (16)$$

which we can rewrite as,

$$\sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} = \frac{\lambda_x E_x(t) + \mu_x N_x}{Z(t)} \quad (17)$$

so, when noticing that, since we are in the large time limit,

$$\sum_{i \in V_x}^{N_x} k = E_x(t) = \lambda_x \langle g_x \rangle \quad (18)$$

and using an identical argument for the x' nodes, we have,

$$\frac{(\lambda_x \langle g_x \rangle + \mu_x f_x)t}{Z(t)} + \frac{(\lambda_{x'} \langle g_{x'} \rangle + \mu_{x'} f_{x'})t}{Z(t)} = 1 \quad (19)$$

$$\implies Z(t) = [(\lambda_x \langle g_x \rangle + \lambda_{x'} \langle g_{x'} \rangle) + (\mu_x f_x + \mu_{x'} f_{x'})]t \quad (20)$$

Which, aside from $\langle g_x \rangle$ and $\langle g_{x'} \rangle$, is an analytical expression for $Z(t)$. For the sake of neat algebra, it is useful to motivate,

$$\tilde{Z} = \frac{m}{(\lambda_x \langle g_x \rangle + \lambda_{x'} \langle g_{x'} \rangle) + (\mu_x f_x + \mu_{x'} f_{x'})} \quad (21)$$

So, if we substitute our newly found values into (6), we get,

$$\begin{aligned} n_x(k, t+1) &= n_x(k, t) + \tilde{Z}t^{-1}(\lambda_x(k-1) + \mu_x)n_x(k-1, t) \\ &\quad - \tilde{Z}t^{-1}(\lambda_x(k) + \mu_x)n_x(k, t) \end{aligned} \quad (22)$$

and since this is the expected number of nodes with an in-degree k at time $t+1$, we can transform this equation into a statement in terms of probabilities,

$$\begin{aligned} (t+1)p_x(k, t+1) - tp_x(k, t) &= \tilde{Z}[(\lambda_x(k-1) + \mu_x)p_x(k-1, t) \\ &\quad - (\lambda_x(k) + \mu_x)p_x(k, t)] + \delta_{k,0} \end{aligned} \quad (23)$$

where we have simply used,

$$p_x(k, t) = \frac{n_x(k, t)}{N_x(t)} \quad (24)$$

Now, we conjecture that the probability of seeing a node of type x for a given in-degree will settle to constant value:

$$\lim_{t \rightarrow \infty} p_x(k, t) = p_{x, \infty}(k) \quad (25)$$

this lets us rewrite (23) as,

$$\frac{p_{x, \infty}(k)}{p_{x, \infty}(k, t)} = \frac{k + \frac{\mu_x}{\lambda_x} - 1}{k + \frac{\mu_x}{\lambda_x} - \frac{1}{\tilde{Z}\lambda_x}} \quad (26)$$

where it is now useful to motivate,

$$\alpha_x = \frac{\mu_x}{\lambda_x} \quad (27)$$

and,

$$\gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x} \quad (28)$$

If we restrict our domain to $k > 0$, we can use the known result that the form of (25) can be written in terms of the ratio of Gamma functions, giving us,

$$p_{x, \infty}(k) = \frac{A_x \Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)} \quad (29)$$

Therefore, if we can solve for the constant A_x , we can predict the probability of a given type of node having a given in-degree. To do this, we consider (26) for $k = 0$,

$$p_{\infty, x}(k = 0) = p_{0, x} = \frac{1}{1 + \tilde{Z}\mu_x} \quad (30)$$

and since (26) is an iterative equation valid when $k = 0$, we can use it to solve for $p_{x, \infty}(k = 1)$, which is within the domain where (29) is a valid. Matching the equation forms and solving for $A(k = 1)$, we not only find that,

$$A(k = 1) = p_{0, x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)} \quad (31)$$

but also conjecture that,

$$A(k) = p_{0,x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)}, \forall k > 0 \quad (32)$$

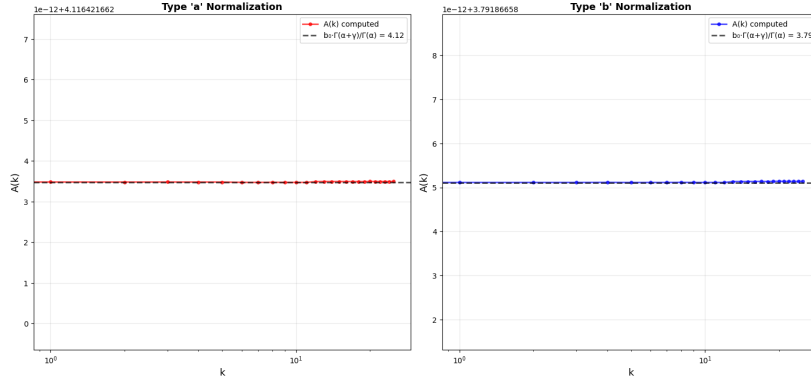


Figure 1: Placeholder Figure. Numerical demonstration of the $A(k) \in \mathbb{R}^+$ conjecture for the integers $k = \{1, 2, \dots, 25\}$. The network model is the simplified version where $x = a$ and $x' = b$, with $f_b = 1 - f_a$ and $h_b = 1 - h_a$. $f_a = 0.4$, $h_a = 0.8$, $\mu_a = 1$, $\mu_b = 2$, $N_{\text{initial}} = 100$, $N_{\text{total}} = N_{\text{initial}} + 25,000$, $m = 40$.

Which is consistent with the known result ((29)). Substituting A_x in gives us,

$$p_{x,\infty}(k) = p_0 \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)} \frac{\Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)} \quad (33)$$

Which is a perfectly valid form to use in our final result however we can simplify further. Using the Beta function, which can be defined as,

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (34)$$

and reintroducing the $k = 0$ case, our equation can be written as,

$$p_{\infty,x}(k) = \begin{cases} p_{0,x} & k = 0 \\ p_{0,x} \cdot \frac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)} & k > 0 \end{cases} \quad (35)$$

and also, by the symmetrical nature of this problem, we have,

$$p_{\infty,x'}(k) = \begin{cases} p_{0,x'} & k = 0 \\ p_{0,x'} \cdot \frac{B(k, \alpha_{x'} + \gamma_{x'})}{B(k, \alpha_{x'})} & k > 0 \end{cases} \quad (36)$$

where now

Therefore, aside from the $\langle g_x \rangle$ and $\langle g_{x'} \rangle$ terms, we now have an analytical solution for the in-degree distribution as $t \rightarrow \infty$, which depends on node type and the levels of homophily and heterophily in the system.

Furthermore, we can reduce these two unknown parameters to one, since, the total number of in-degrees in the network added at each timestep needs to equal the total number of out-degrees added at each timestep. In other words,

$$\langle g_x \rangle + \langle g_{x'} \rangle = m \quad (37)$$

Numerically, we see not only do the proportion of in-edges per node type tend to constant values as $t \rightarrow \infty$, but these constants also sum to m .

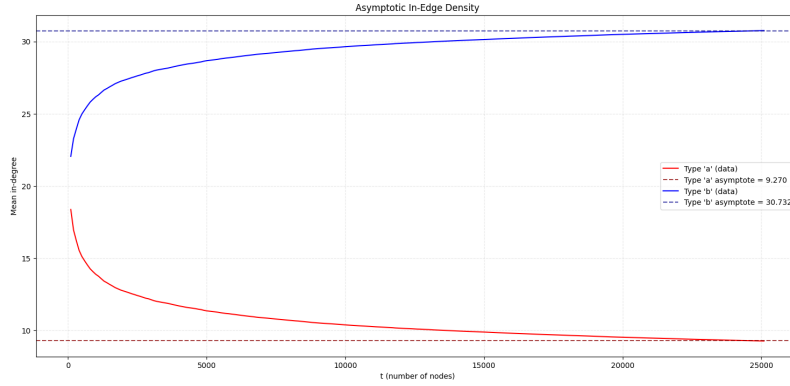


Figure 2: Placeholder Figure. Numerical demonstration of the proportion of in-edges per node type tending to constant values as $t \rightarrow \infty$. This also shows these asymptotes, as expected, sum to m . The network model is the simplified version where $x = a$ and $x' = b$, with $f_b = 1 - f_a$ and $h_b = 1 - h_a$. $f_a = 0.4$, $h_a = 0.8$, $\mu_a = 1$, $\mu_b = 2$, $N_{\text{initial}} = 100$, $N_{\text{total}} = N_{\text{initial}} + 25,000$, $m = 40$.

5 Numerical Model

One way to assess whether our theory remains valid for a finite-sized preferential attachment network with homophily is to numerically simulate the network and compare the per-type in-degree probability distributions with the theoretical predictions in Eqs.(35) and (36). To observe homophilic preferential attachment, an initial network is required, since at each time step the attachment process

necessitates an existing target node; without such a target, the notion of in-degree is undefined. We are therefore tasked with setting up this initial network such that choices made regarding the initial set up are made negligible in the long time limit.

One might think that the least intrusive choice of initialisation is a topology that looks as if it itself has been created by a heterophilic preferential attachment process with the same exogenous system parameters we are attempting to model. Let us set up what we can call our *minimally-intrusive initialisation algorithm*,

We start our model with an initialisation fraction f_x of x type nodes (meaning the remaining fraction of nodes is $f_{x'}$). If we construct this model probabilistically (i.e. the algorithms actualisations are dependent on whether the probabilities outlined above fall above or below a numpy randomly generated number in $[0, 1]$), clearly, we have $N_{initial}$ nodes with an expected,

$$\mathbb{E}[N_{x,initial}] = f_x N_{initial} \quad (38)$$

and,

$$\mathbb{E}[N_{x',initial}] = f_{x'} N_{initial} \quad (39)$$

For simplicity let,

$$t_{initial} = N_{initial} \quad (40)$$

The next step is to randomly select a source node, which then connects to a target node chosen according to a specified probability distribution. For example, if we let the x type nodes have a probability of forming a connection with a node another node in the initial system,

$$\Pi_x(k, t) = \frac{\lambda_x k + \mu_x}{Y(t)}, t \in t_{initial} \quad (41)$$

To find $Y(t)$, we follow a derivation similar to that of $Z(t)$ found earlier. There are two key differences here however; firstly, since the source node is already a member of the network, we need to explicitly specify that the node cannot form a connection with itself, and secondly, we cannot use the asymptotic convergence trick used in equation (14) to find a linear constant as the in-degree per node-type is non-constant in $t \in t_{initial}$. To account for this first discrepancy, we can simply add a step in our algorithm that checks the type and in-degree of the randomly selected node and then removes the associated contributions this would give to $Y(t)$ were it included. For example, if the randomly selected node is of type x and has a total in-degree of k_i (where i simply labels the particular

node in the system we are considering) at time $t \in t_{initial}$, then $Y(t)$ can be written as,

$$Y(t) = \lambda_x(E_x - k_i) + (N_x - 1)\mu_x + N_{x'}\mu_{x'} \quad (42)$$

a reasonable choice for the total edge counts is exogenous parameter informed expected values i.e,

$$E_{total}^{(in)} = \mathbb{E}[E_x(t)] + \mathbb{E}[E_{x'}(t)] = \lambda_x m t + \lambda_{x'} m t, t \in t_{initial} \quad (43)$$

However this initialisation has a drawback: since $t_{initial} < \infty$, the realised values will not necessarily match the expected. If we instead used a deterministic initialisation matching expected values exactly, we would be limited to parameters resulting in integer expectations. This constraint would severely restrict the model's application potential. What we conclude from this is that an initialisation which appears exactly as one would expect from a heterophilic preferential attachment process either does not align perfectly or is limited in its range of applicability. As it turns out, so long as,

$$\lim_{t \rightarrow \infty} N_{initial}/N(t) = 0 \quad (44)$$

The effect of the initial conditions are negligible [4]. motivates the question, *"Does this result hold on a finite-sized network?"*

- Research more literature on finite model initial conditions
- Demonstrate that empirically these conditions do wash out for a few initialisation choices, e.g. the one described above, ER, and the one i use in my model.
- outline the algorithm and any other numerical discrepancies from theory (there aren't many in the theory, the tricky part is in the computational statistical tests. This work is ongoing.

6 Statistical Analysis of Numerical Model

- Show that Sterlings approximation is valid on the data
- Show that the resulting power law is/is not an appropriate fit. for what parameter choices does this hold for?
- show the beta fits the data better.

- show the improved Clauset is one way to go. find if there is active research in this field. Outline that, if there is, why we have chosen the test we have.
- Try and justify why this test is also valid for the beta dist. If you cant justify then admit then conjecturing this statistical test is our best case scenario is the way to go.

The Cumulative Distribution Function for this is:

$$F_{\infty,x}(k) = \int_{-\infty}^k p_{\infty,x}(t) dt \quad (45)$$

But it appears the analytical solution to this is not a known result.

- KS test issues. Not sure if our semi analytic result allows for use of KS test.
- investigate why the KS isnt working in terms of how the p-val is calculated
- read the paper on the tails using ks tests
- Demonstrate empirical exploration of the coupling (non-independence between nodes) renders the reduced chi-squared inappropriate.
- Discuss chi-squared Pearson assumption in Scipy and why this doesn't hold under strong coupling.
- outline theoretically that Poisson distribution per bin is theoretically not followed.
- examine suitability of monte carlo under the assumptions that allow for chi-sq
- outline conditions where approximations hold investigate for a series of conjectured underlying distributions and see if any approximately hold (they probably wont but will make for nice analysis)

7 Misc

References

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